Chapter 8. Comparing Estimators

Criteria, sufficiency, UMVUE, asymptotic efficiency

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8.1 Criteria for comparing estimators

For an estimator $\hat{\eta}$ of $\eta = \eta(\theta)$, the **mean squared error** is

$$\mathrm{MSE}(\hat{\eta}, \theta) = E_\theta(\hat{\eta} - \eta)^2 = \mathrm{Var}_\theta\, \hat{\eta} + \{\mathrm{bias}_\theta(\hat{\eta})\}^2.$$

No estimator minimizes the MSE uniformly in θ: the constant $\hat{\eta} \equiv \eta(\theta_0)$ has MSE zero at θ_0. So one restricts the class, most commonly to **unbiased** estimators ($E_\theta \hat{\eta} = \eta$ for all θ), and seeks the **uniformly minimum variance unbiased estimator** (UMVUE).

Example 8.1

For $N(\mu, \sigma^2)$, $S^2 = \frac{1}{n-1}\sum (X_i - \bar{X})^2$ is unbiased with MSE $= \frac{2\sigma^4}{n-1}$. The MLE $\hat{\sigma}^2 = \frac{n-1}{n} S^2$ has MSE $= \frac{(2n-1)\sigma^4}{n^2}$, which is smaller. Unbiasedness is a restriction, not a guarantee of small error.

8.2 Sufficient statistics

Definition (Sufficiency)

A statistic $Y = u(\mathbf{X})$ is **sufficient** for θ if the conditional distribution of $\mathbf{X}$ given $Y = y$ does not depend on θ.

Once Y is known, the rest of the sample carries no information about θ. For Bernoulli(θ), $P_\theta(\mathbf{X} = \mathbf{x} \mid \sum X_i = y) = \binom{n}{y}^{-1}$ whenever $\sum x_i = y$, so $\sum X_i$ is sufficient.

Theorem 8.1 (Factorization)

$Y = u(\mathbf{X})$ is sufficient iff $\prod_i f(x_i; \theta) = k_1(u(\mathbf{x}), \theta)\, k_2(\mathbf{x})$ for some functions k_1, k_2.

Proof (discrete case). If the factorization holds, then for $u(\mathbf{x}) = y$,

$$P_\theta(\mathbf{X} = \mathbf{x} \mid Y = y) = \frac{k_1(y, \theta) k_2(\mathbf{x})}{\sum_{\mathbf{z}: u(\mathbf{z}) = y} k_1(y, \theta) k_2(\mathbf{z})} = \frac{k_2(\mathbf{x})}{\sum_{u(\mathbf{z}) = y} k_2(\mathbf{z})},$$

which is free of θ. Conversely, if Y is sufficient, $P_\theta(\mathbf{X} = \mathbf{x}) = P(\mathbf{X} = \mathbf{x} \mid Y = u(\mathbf{x}))\, P_\theta(Y = u(\mathbf{x}))$, which is the factorization with $k_2(\mathbf{x}) = P(\mathbf{X} = \mathbf{x} \mid Y = u(\mathbf{x}))$. ■

Examples: $\sum X_i$ for Poisson and exponential; $(\sum X_i, \sum X_i^2)$ for $N(\mu, \sigma^2)$; $X_{(n)}$ for $U(0, \theta)$, since

$$\textstyle\prod_i \theta^{-1} I(0 < x_i < \theta) = \theta^{-n} I(x_{(n)} < \theta)\, I(x_{(1)} > 0).$$

For a k-parameter exponential family $f(x; \theta) = \exp\{\sum_j \eta_j(\theta) T_j(x) - B(\theta) + S(x)\}$, the vector of sums $\sum_i T_j(X_i)$, $j = 1, \ldots, k$, is sufficient.

Theorem 8.2 (Rao–Blackwell)

Let Y be sufficient and $\hat{\eta}$ any unbiased estimator of η. Then $\hat{\eta}^{RB} = E(\hat{\eta} \mid Y)$ is a statistic, is unbiased, and $\mathrm{Var}_\theta\, \hat{\eta}^{RB} \le \mathrm{Var}_\theta\, \hat{\eta}$ for all θ.

Proof. By sufficiency the conditional expectation does not involve θ, so it is a statistic. Unbiasedness follows from the tower property. The variance decomposition gives $\mathrm{Var}\, \hat{\eta} = E\, \mathrm{Var}(\hat{\eta} \mid Y) + \mathrm{Var}\, E(\hat{\eta} \mid Y) \ge \mathrm{Var}\, \hat{\eta}^{RB}$. ■

So we need only look among functions of a sufficient statistic.

8.3 Complete sufficient statistics and the UMVUE

Definition (Completeness)

Y is **complete** if $E_\theta g(Y) = 0$ for all $\theta \in \Omega$ implies $P_\theta(g(Y) = 0) = 1$ for all θ.

Theorem 8.3 (Lehmann–Scheffé)

If Y is complete sufficient, then (a) any unbiased estimator of the form $\delta(Y)$ is the (essentially unique) UMVUE; (b) equivalently, $E(\hat\eta \mid Y)$ is the UMVUE for every unbiased $\hat\eta$.

Proof. Let $\delta_1(Y), \delta_2(Y)$ both be unbiased. Then $E_\theta\{\delta_1(Y) - \delta_2(Y)\} = 0$ for all θ, so $\delta_1 = \delta_2$ a.s. by completeness. Given any unbiased $\hat\eta$, Rao–Blackwell gives the unbiased function $E(\hat\eta \mid Y)$ of Y with no larger variance, and by uniqueness it equals $\delta(Y)$. So $\delta(Y)$ has variance at most that of every unbiased estimator. ∎

- **Binomial:** for $Y \sim B(n,\theta)$, $\sum_y g(y)\binom{n}{y}\lambda^y = 0$ for all $\lambda = \theta/(1-\theta) > 0$ forces every coefficient to vanish (a polynomial identity). So Y is complete and $\bar X$ is the UMVUE of θ.
- **Poisson:** the same argument with a power series in $\lambda = n\theta$.
- **Exponential:** $Y = \sum X_i \sim \text{Gamma}(n,\theta)$, and uniqueness of the Laplace transform gives completeness.
- **Exponential families:** if the natural parameter space contains a k-dimensional open set, $(\sum_i T_1, \dots, \sum_i T_k)$ is complete. Hence $\bar X$ and S^2 are UMVUEs for a normal sample.

Example 8.2

For Poisson(θ) and $\eta = P(X = 0) = e^{-\theta}$, the indicator $I(X_1 = 0)$ is unbiased. Given $Y = \sum X_i = y$, $X_1 \sim B(y, 1/n)$, so the UMVUE is $E\{I(X_1 = 0) \mid Y\} = (1 - \frac{1}{n})^Y$. (The MLE $e^{-\bar X}$ is biased.)

Example 8.3

For $U(0,\theta)$, $Y = X_{(n)}$ has density ny^{n-1}/θ^n. If $\int_0^\theta g(y)y^{n-1}\,dy = 0$ for all θ, differentiating gives $g \equiv 0$, so Y is complete. Since $EY = \frac{n}{n+1}\theta$, the UMVUE is $\frac{n+1}{n}X_{(n)}$.

The information inequality.

Theorem 8.4 (Cramér–Rao)

Under the regularity conditions of Lemma 6.4, if $\hat\eta$ is unbiased for $\eta(\theta)$, with η differentiable, then $\text{Var}_\theta\,\hat\eta \geq \dfrac{\eta'(\theta)^2}{nI(\theta)}$.

Proof. Differentiate $\int \hat\eta(\mathbf{x}) \prod f(x_i;\theta)\,d\mathbf{x} = \eta(\theta)$ under the integral sign: $\text{Cov}_\theta(\hat\eta, \dot l_n(\theta)) = \eta'(\theta)$, since $E\dot l_n = 0$. By Cauchy–Schwarz, $\eta'(\theta)^2 \leq \text{Var}\,\hat\eta \cdot \text{Var}\,\dot l_n = \text{Var}\,\hat\eta \cdot nI(\theta)$. ∎

Equality holds iff $\dot l_n(\theta)$ is linear in $\hat\eta$, i.e. essentially only in exponential families with $\hat\eta$ the mean of the natural statistic. The bound can be unattainable even when a UMVUE exists, as in Example 8.2.

8.4 Asymptotic efficiency

Definition (Asymptotic relative efficiency)

If $\sqrt{n}(\hat{\theta}_n^{(i)} - \theta) \xrightarrow{d} N(0, \sigma_i^2(\theta))$ for $i = 1, 2$, then $ARE(\hat{\theta}^{(1)}, \hat{\theta}^{(2)}) = \sigma_2^2(\theta)/\sigma_1^2(\theta)$.

The ARE is the limiting ratio of sample sizes needed for the same precision: estimator 2 needs ARE times as many observations as estimator 1 for intervals of the same length. By Theorem 6.6 the MLE attains asymptotic variance $1/I(\theta)$. Under regularity no reasonable estimator does better, so the MLE is **asymptotically efficient**.

Example 8.4 (Location models)

For $f(x-\theta)$ with f symmetric, the mean has asymptotic variance σ^2, the median $1/(4f(0)^2)$ (Theorem 5.8 in Volume I), and the MLE $1/I$. With the standard logistic density, $\sigma^2 = \pi^2/3$, $f(0) = \tfrac{1}{4}$ and $I = \tfrac{1}{3}$. So the asymptotic variances are 3 (MLE), 3.29 (mean) and 4 (median), and $ARE(\text{mean}, \text{median}) = 12/\pi^2 \approx 1.22$. For the normal, $ARE(\text{mean}, \text{median}) = \pi/2$; for the Laplace, the median is the MLE and beats the mean by a factor of 2.